3 Configuration

Tidewater reads its settings from a single TOML file when the server starts. Most settings can also be overridden with environment variables, which is convenient in container deployments.

3.1 Configuration file location

By default the server looks for `/etc/tidewater/tidewater.toml`. You can point to another file with the `--config` option:

```
tidewater-server --config /opt/tidewater/conf/prod.toml
```

3.2 A minimal configuration

The following file is enough to run a single server with token authentication:

```
[server]
bind = "0.0.0.0:7420"
data_dir = "/var/lib/tidewater"
max_connections = 2048

[storage]
segment_size_mb = 256
fsync = "batch"
retention_hours = 168

[auth]
mode = "token"
token_file = "/etc/tidewater/tokens.json"
```

Note – Changes to the [storage] section take effect only after a full restart; a reload signal is not enough.

3.3 Parameter reference

Parameter	Type	Default	Description
`server.bind`	string	`"127.0.0.1:7420"`	Address and port on which the server accepts client connections.
`server.data_dir`	path	`/var/lib/tidewater`	Directory for queue segments and metadata.
`server.max_connections`	integer	`1024`	Maximum number of simultaneous client connections.
`server.idle_timeout_s`	integer	`300`	Seconds after which an idle connection is closed.
`storage.segment_size_mb`	integer	`128`	Size at which a segment file is rolled over.
`storage.fsync`	string	`"interval"`	When data is flushed to disk: "always", "batch" or "interval".
`storage.fsync_interval_ms`	integer	`200`	Flush interval when fsync is "interval".
`storage.retention_hours`	integer	`72`	How long acknowledged messages are kept.
`storage.compression`	string	`"none"`	Segment compression: "none", "lz4" or "zstd".
`auth.mode`	string	`"none"`	Client authentication: "none", "token" or "mtls".
`auth.token_file`	path	`(none)`	JSON file holding the accepted client tokens.
`tls.cert_file`	path	`(none)`	Server certificate in PEM format.
`tls.key_file`	path	`(none)`	Private key for the server certificate.

Parameter	Type	Default	Description
`metrics.enabled`	boolean	`true`	Exposes server metrics in plain text over HTTP.
`metrics.bind`	string	`"127.0.0.1:9420"`	Address of the metrics endpoint.
`log.level`	string	`"info"`	One of "error", "warn", "info", "debug" or "trace".

3.4 Environment variable overrides

Any parameter can be set through an environment variable whose name is TIDEWATER_ followed by the section and the key in capital letters. For example:

```
export TIDEWATER_SERVER_MAX_CONNECTIONS=4096
export TIDEWATER_STORAGE_FSYNC=always
export TIDEWATER_LOG_LEVEL=debug
```

Environment variables take precedence over values in the configuration file.

3.5 Validating a configuration

Before restarting a production server, check the file with the built-in validator:

```
$ tidewater-server check --config /etc/tidewater/tidewater.toml
config OK: 14 parameters set, 0 warnings
```

The validator exits with status 1 if it finds an unknown parameter or a value of the wrong type.